THE TROTTER EXAM SOLUTIONS

ZIMBABWE SCHOOLS EXAMINATION COUNCIL
General Certificate Of Education Advanced Level

PURE MATHEMATICS **6042/1**
PAPER 1

NOVEMBER 2021 SUGGESTED MARKING GUIDE BY

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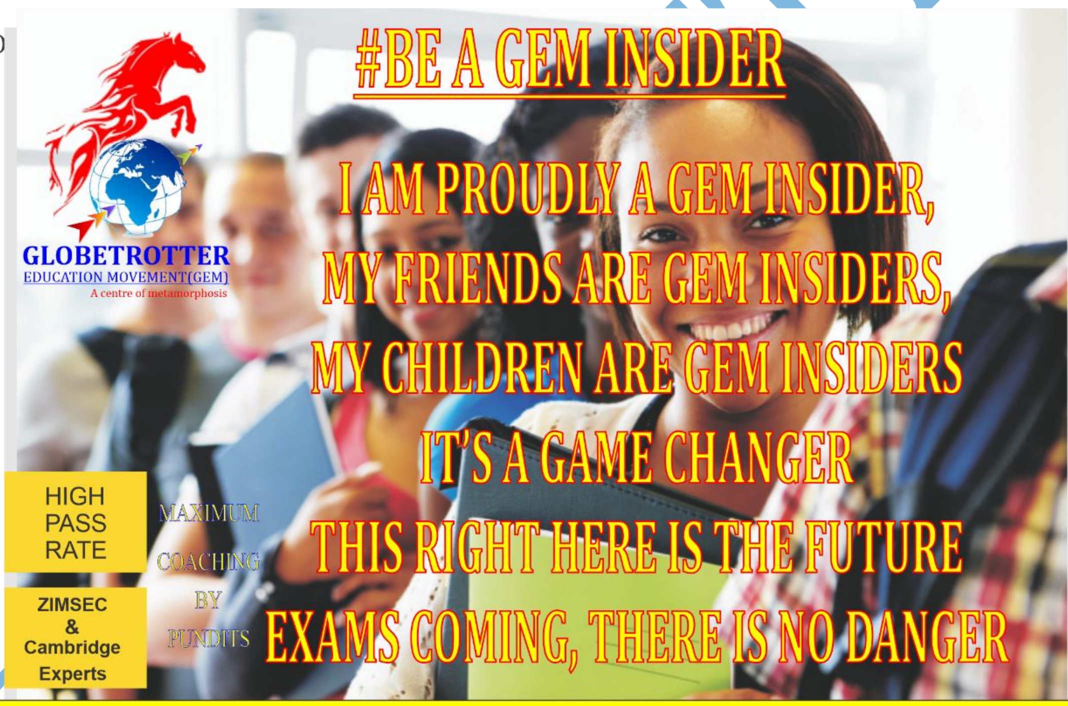

This marking guide consists of 27 printed pages.

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VOICE OF THE PROFESSOR (VOP)

1. $2e^{2x} - 7e^{x} + 6 = 0$

$$let\ y = e^{x}$$
$$2y^2 - 27y + 6 = 0$$
$$2y^2 - 4y - 3y + 6 = 0$$
$$2y(y-2) - 3(y-2) = 0$$
$$(y-2)(2y-3) = 0$$
$$y = 2\ or \frac{3}{2}$$

$$[\because y = e^{x}]$$

$$\therefore e^{x} = 2$$
$$\ln e^{x} = \ln 2$$
$$\boldsymbol{x = \ln 2}$$

OR

$$e^{x} = \frac{3}{2}$$
$$\ln e^{x} = \ln\left(\frac{3}{2}\right)$$
$$\boldsymbol{x = \ln\left(\frac{3}{2}\right)}$$

[Award 3 marks]

2. $\frac{\sin 2\theta}{1+\cos 2\theta} = \tan\theta$

$$LHS = \frac{sin\ 2\theta}{1 + cos\ 2\theta}$$

$apply\ the\ double\ angle\ formulae$

$\sin 2\theta = 2\ sin\ \theta\ cos\ \theta$

$\cos 2\theta = 2\cos^2\theta - 1$

$Or\ \cos 2\theta = 1 - 2\sin^2\theta$

$$= \frac{2\sin\theta\cos\theta}{1 + (1 - 2\sin^2\theta)}$$

$$= \frac{2\sin\theta\cos\theta}{2 - 2\sin^2\theta}$$

$$= \frac{2\ sin\ \theta\ cos\ \theta}{2(1 - sin^2\ \theta)}$$

$$= \frac{2\ sin\ \theta\ cos\ \theta}{2(\cos^2\theta)}$$

$$= \frac{\sin\theta}{\cos\theta}$$

$$\boldsymbol{= \tan\theta}$$
$$\boldsymbol{LHS = RHS\ \ (proved)}$$

[Award 3 marks]

Alternative solution

$$\frac{\sin 2\theta}{1+\cos 2\theta} = \tan\theta$$

$$LHS = \frac{\sin 2\theta}{1+\cos 2\theta}$$

Apply the double angle formulae

We know that

$\sin 2\theta = 2\sin\theta\cos\theta$

and

$\cos 2\theta = 2\cos^2\theta - 1$

$\therefore$

$1+\cos 2\theta = 2\cos^2\theta$

Now

$$LHS = \frac{2\sin\theta\cos\theta}{2\cos^2\theta}$$

$$= \frac{2\sin\theta\cos\theta}{2\cos\theta\cos\theta}$$

$$= \frac{\sin\theta}{\cos\theta}$$

$$= \mathbf{\tan\theta}$$

$$\boldsymbol{LHS = RHS} \quad (\boldsymbol{proved})$$

VOICE OF THE PROFESSOR (VOP)

Key Identities

$\sin 2\theta = 2\sin\theta\cos\theta$

and

$\cos 2\theta = 2\cos^2\theta - 1$

$\therefore$

$1+\cos 2\theta = 2\cos^2\theta$

Accept ORA

[Award 3 marks]

3. $y = \ln[\cos(x^2+1)]$

$$\frac{dy}{dx} = \frac{\frac{d}{dx}[\cos(x^2+1)]}{\cos(x^2+1)}$$

$$= \frac{-\sin(x^2+1)\cdot\frac{d}{dx}(x^2+1)}{\cos(x^2+1)}$$

$$= \frac{-\sin(x^2+1)\cdot 2x}{\cos(x^2+1)}$$

$$= \mathbf{-2x\tan(x^2+1)}$$

as required

Recall

If $y = \ln[f(x)]$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

Accept ORA

[Award 3 marks]

4. (a) $M \propto \frac{1}{\sqrt[3]{n-1}}$

$$M = \frac{k}{\sqrt[3]{n-1}}$$

when $M = 5, n = 9$

 Important message: Jesus is coming again

To God be the glory

VOICE OF THE PROFESSOR (VOP)

$$5 = \frac{k}{\sqrt[3]{9-1}}$$

$$5 = \frac{k}{\sqrt[3]{8}}$$

$$5 = \frac{k}{2}$$

$$10 = k$$

$$\therefore \textbf{The formula is } M = \frac{\mathbf{10}}{\sqrt[3]{\mathbf{n-1}}}$$

[Award 3 marks]

(b) $M = \frac{10}{\sqrt[3]{n-1}}$

$$when\ M = 25$$

$$25 = \frac{10}{\sqrt[3]{n-1}}$$

$$25\sqrt[3]{n-1} = 10$$

$$\sqrt[3]{n-1} = \frac{10}{25}$$

$$\sqrt[3]{n-1} = \frac{2}{5}$$

$$n-1 = \left(\frac{2}{5}\right)^3$$

$$n-1 = \frac{8}{125}$$

$$n = \frac{8}{125} + 1$$

$$\boldsymbol{n = \frac{133}{125}}$$

1.064 *and* $1\frac{8}{125}$ *are also acceptable equivalent exact values*

[Award 2 marks]

5. **(a)** $2x^2 + 3x + 1 \equiv a(x+b)^2 + c$

$$2x^2 + 3x + 1 \equiv a(x^2 + 2bx + b^2) + c$$

$$2x^2 + 3x + 1 \equiv ax^2 + 2abx + ab^2 + c$$

comparing coefficients

VOICE OF THE PROFESSOR (VOP)

$$x^2 \to 2 = a$$

$$x \to 3 = 2ab$$

$$3 = 2(2)(b)$$

$$3 = 4b$$

$$\frac{3}{4} = b$$

$$constants \to 1 = ab^2 + c$$

$$1 = 2\left(\frac{3}{4}\right)^2 + c$$

$$1 = \frac{9}{8} + c$$

$$1 - \frac{9}{8} = c$$

$$-\frac{\mathbf{1}}{\mathbf{8}} = \boldsymbol{c}$$

$$\therefore \boldsymbol{f(x) = 2\left(x + \frac{3}{4}\right)^2 - \frac{1}{8}}$$

[Award 3 marks]

Alternative solution

$$\therefore \boldsymbol{f(x) = 2\left(x + \frac{3}{4}\right)^2 - \frac{1}{8}}$$

$$\boldsymbol{where\ a = 2, b = \frac{3}{4}\ and\ c = -\frac{1}{8}}$$

Alternative solution

$$let\ f(x) = 2x^2 + 3x + 1$$

$$= 2\left[x^2 + \frac{3}{2}x + \frac{1}{2}\right]$$

Make sure that the coefficient of x^2 is $+1$

$$= 2\left[x^2 + \frac{3}{2}x + \left(\frac{3}{2}\right)^2 + \frac{1}{2} - \left(\frac{3}{2}\right)^2\right]$$

$$= 2\left[\left(x + \frac{3}{4}\right)^2 + \frac{1}{2} - \frac{9}{16}\right]$$

$$= 2\left[\left(x + \frac{3}{4}\right)^2 - \frac{1}{16}\right]$$

$$= \boldsymbol{2\left(x + \frac{3}{4}\right)^2 - \frac{1}{8}}$$

[Award 3 marks]

$$\boldsymbol{where\ a = 2, \quad b = \frac{3}{4} \quad and \quad c = -\frac{1}{8}}$$

VOICE OF THE PROFESSOR (VOP)

(b) $2x^2 + 3x + 1 = 0$

hence

$$2\left(x + \frac{3}{4}\right)^2 - \frac{1}{8} = 0$$

$$2\left(x + \frac{3}{4}\right)^2 = \frac{1}{8}$$

$$\left(x + \frac{3}{4}\right)^2 = \frac{1}{8 \times 2}$$

$$\left(x + \frac{3}{4}\right)^2 = \frac{1}{16}$$

$$x + \frac{3}{4} = \sqrt{\frac{1}{16}}$$

$$x + \frac{3}{4} = \pm\frac{1}{4}$$

$$x = -\frac{3}{4} \pm \frac{1}{4}$$

$$\boldsymbol{x = -1\ or -\frac{1}{2}}$$

[Award 3 marks]

Alternative solution (Otherwise)

$$2x^2 + 3x + 1 = 0$$

$$2x^2 + 2x + x + 1 = 0$$

$$2x(x + 1) + 1(x + 1) = 0$$

$$(x + 1)(2x + 1) = 0$$

$$\boldsymbol{x = -1\ or -\frac{1}{2}}$$

[Award 3 Marks]

VOICE OF THE PROFESSOR (VOP)

Alternative solution (Otherwise)

$$2x^2 + 3x + 1 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-3 \pm \sqrt{3^2 - 4(2)(1)}}{2(2)}$$

$$x = \frac{-3 \pm \sqrt{9 - 8}}{4}$$

$$x = \frac{-3 \pm \sqrt{1}}{4}$$

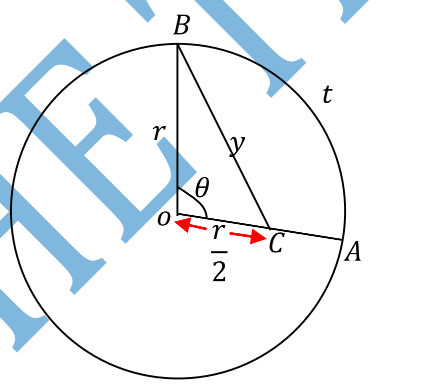
$$x = \frac{-3 \pm 1}{4}$$

$$x = \frac{-3 + 1}{4} \ or \ x = \frac{-3 - 1}{4}$$

$$\boldsymbol{x = -\frac{1}{2} \ or -1}$$

[Award 3 Marks]

6. (a)



Nota Bene:
In plane trigonometry, it is crucial to add annotations to your diagram

Since C is the midpoint of OA, $OC = \frac{r}{2}$

Using the cosine rule

$$y^2 = r^2 + \left(\frac{r}{2}\right)^2 - 2r\left(\frac{r}{2}\right)\cos\theta$$

$$y = r^2 + \frac{r^2}{4} - r^2\cos\theta$$

$$y^2 = \frac{4r^2 + r^2}{4} - r^2\cos\theta$$

$$y^2 = \frac{5r^2}{4} - r^2\cos\theta$$

[Award 2 Marks]

VOICE OF THE PROFESSOR (VOP)

Key Formulae

If θ is small, $\cos\theta \approx 1 - \frac{\theta^2}{2}$

Arc length $= r\theta$

(b) $y^2 = \frac{5r^2}{4} - r^2\cos\theta$

Aside

(b) $y^2 = \frac{5r^2}{4} - r^2\cos\theta$

If θ is small, $\cos\theta \approx 1 - \frac{\theta^2}{2}$

$$\Rightarrow y^2 \approx \frac{5r^2}{4} - r^2\left(1 - \frac{\theta^2}{2}\right)$$

$$y^2 \approx \frac{5r^2}{4} - r^2 - \frac{r^2\theta^2}{2}$$

we need to eliminate θ and introduce t

Aside

we know that arc length $= r\theta$

$\therefore t = r\theta$

$\frac{t}{r} = \theta$

Now substituting

$$y^2 \approx \frac{5r^2}{4} - r^2 - \frac{r^2\left(\frac{t}{r}\right)^2}{2}$$

$$y^2 \approx \frac{5r^2 - 4r^2}{4} - \frac{r^2\left(\frac{t^2}{r^2}\right)}{2}$$

$$y^2 \approx \frac{r^2}{4} - \frac{t^2}{2}$$

As required

[Award 4 Marks]

7. $A = \begin{pmatrix} 3 & -1 \\ 4 & -1 \end{pmatrix}$

$$A^n = \begin{pmatrix} 2n+1 & -n \\ 4n & 1-2n \end{pmatrix}$$

Intial stage: when $n = 1$

$$LHS = A^n$$

$$= A^1$$

$$= \begin{pmatrix} 3 & -1 \\ 4 & -1 \end{pmatrix}^1$$

$$= \begin{pmatrix} 3 & -1 \\ 4 & -1 \end{pmatrix}$$

$$RHS = \begin{pmatrix} 2n+1 & -n \\ 4n & 1-2n \end{pmatrix}$$

$$= \begin{pmatrix} 2(1)+1 & -1 \\ 4(1) & 1-2(1) \end{pmatrix}$$

$$= \begin{pmatrix} 3 & -1 \\ 4 & -1 \end{pmatrix}$$

$LHS = RHS$

$hence\ true\ for\ n = 1$

$Assumption\ stage$

$assume\ true\ for\ n = k+1$

$$A^k = \begin{pmatrix} 2k+1 & -k \\ 4k & 1-2k \end{pmatrix}$$

$Thesis$

$when\ n = k+1$

$$A^{k+1} = \begin{pmatrix} 2(k+1)+1 & -(k+1) \\ 4(k+1) & 1-2(k+1) \end{pmatrix}$$

$Proof:$

$LHS = A^{k+1}$

$Applying\ laws\ of\ indices$

$= A^k \cdot A^1$

$by\ the\ inductive\ hypothesis$

$$= \begin{pmatrix} 2k+1 & -k \\ 4k & 1-2k \end{pmatrix}\begin{pmatrix} 3 & -1 \\ 4 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 6k+3-4k & -2k-1+k \\ 12k+4-8k & -4k-1+2k \end{pmatrix}$$

$$= \begin{pmatrix} 2k+3 & -k-1 \\ 4k+4 & -2k-1 \end{pmatrix}$$

$creating\ a\ situation\ and\ factorising$

$$= \begin{pmatrix} 2k+2+1 & -(k+1) \\ 4(k+1) & -2k-2+1 \end{pmatrix}$$

$$= \begin{pmatrix} 2(k+1)+1 & -(k+1) \\ 4(k+1) & 1-2(k+1) \end{pmatrix}$$

$LHS = RHS, hence\ true\ for\ n = k+1$

$conclusion$

$\boldsymbol{Since\ the\ statement\ is\ true\ for\ n = 1, n = k, n = k+1,}$

$\boldsymbol{\therefore\ by\ induction, the\ statement\ holds\ for\ all\ \ n \in \mathbb{Z}^+}$

VOICE OF THE PROFESSOR (VOP)

Nota Bene:

On the proof stage, the thesis stage marries the assumption stage.

In this case, the, take LHS of the Thesis and apply laws of indices and substitute the RHS of the assumption

Then create a situation with a bias towards the RHS of the Thesis.

[Award 7 marks]

VOICE OF THE PROFESSOR (VOP)

8. (a) $\frac{2x^2+1}{2^x-1}=5$

$$let\ y=2^x$$
$$\frac{y+1}{y-1}=5$$
$$y+1=5(y-1)$$
$$y+1=5y-5$$
$$1+5=5y-y$$
$$6=4y$$
$$y=\frac{3}{2}$$
$$\because y=2^x$$
$$\therefore 2^x=\frac{3}{2}$$
$$\ln 2^x=\ln\left(\frac{3}{2}\right)$$
$$x\ln 2=\ln\left(\frac{3}{2}\right)$$
$$x=\frac{\ln\left(\frac{3}{2}\right)}{\ln 2}$$
$$x=0.584962500$$
$$\boldsymbol{x=0.585\ to\ 3sf}$$

[Award 3marks]

(b) $|x-3|>2|3x+1|$

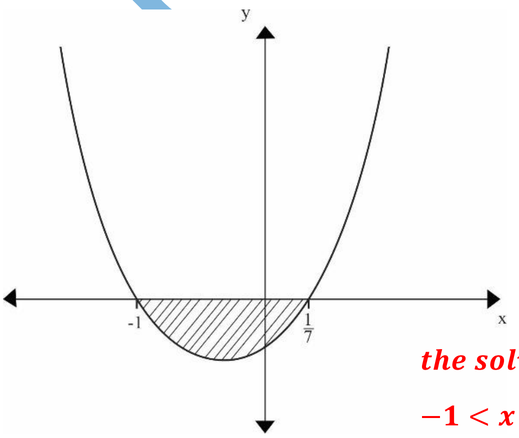
squaring both sides

$$(x-3)^2>2^2(3x+1)^2$$
$$x^2-6x+9>4(9x^2+6x+1)$$
$$x^2-6x+9>36x^2+24x+4$$
$$0>36x^2-x^2+24+6x+4-9$$
$$0>35x^2+30x-5$$
$$0>7x^2+6x-1$$
$$7x^2+6x-1<0$$
$$7x^2+7x-x-1<0$$
$$(7x-1)(x+1)<0$$

critical values are

$$x=-1\ \ and\ x=\frac{1}{7}$$

If the modulus is on both sides, square both sides and proceed as usual.



A table could also be used

the solution is

$$\boldsymbol{-1<x<\frac{1}{7}}$$

[Award 5 marks]

Alternative solution

$$|x-3| > 2|3x+1|$$

$$|x-3| > |6x+2|$$

square both sides

$$(x-3)^2 > (6x+2)^2$$

$rearrange\ terms\ withe\ a\ bias\ towards\ difference\ of\ 2\ squares$

$$(x-3)^2 > (6x+2)^2$$

$$(x-3)^2 - (6x+2)^2 > 0$$

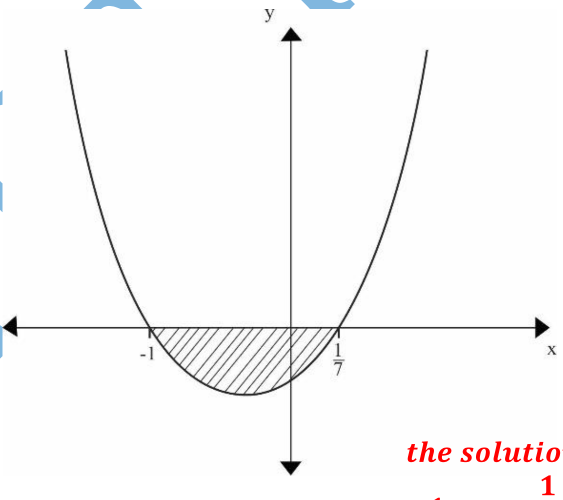
$factorise\ LHS\ using\ difference\ of\ two\ squares$

$$(x-3+6x+2)(x-3-6x-2) > 0$$

$$(7x-1)(-5x-5) > 0$$

$crititical\ values\ are$

$$x = -1 \ and\ x = \frac{1}{7}$$



the solution is

$$\mathbf{-1 < x < \frac{1}{7}}$$

VOICE OF THE PROFESSOR (VOP)

If the modulus is on both sides, square both sides and proceed as usual.

A table could also be used

[Award 5 marks]

9. (a) $\frac{e^x}{1+x}$

$= e^x(1+x)^{-1}$

Write down the expansion for e^x from the MF7 formula list

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}$$

$$= 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$$

find the expansion of $(1+x)^{-1}$ using bionomial theorem

$$(1+x)^n = 1 + \frac{nx}{1!} + \frac{n(n-1)x^2}{2!} + \frac{n(n-1)(n-2)x^3}{3!}$$

$$(1+x)^{-1} = 1 + (-1)x + \frac{(-1)(-2)x^2}{3\times 2} + \frac{(-1)(-2)(-3)x^3}{3\times 2}$$

$$= 1 - x + x^2 - x^3$$

Now

$$e^x(1+x)^{-1} = \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6}\right)(1 - x + x^2 - x^3)$$

expand the brackets, ignoring all terms in x^4 and higher

$$= 1 - x + x^2 - x^3 + x - x^2 + x^3 + \frac{x^2}{2} - \frac{x^3}{2} + \frac{x^3}{6}$$

group like terms

$$= 1 - x + x + x^2 - x^2 + \frac{x^2}{2} - x^3 + x^3 - \frac{x^3}{2} + \frac{x^3}{6}$$

$$= \mathbf{1 + \frac{x^2}{2} + \frac{x^3}{3}} \qquad \textbf{\textit{as required}}$$

(b) $\frac{e^x}{1+x} = 1 + \frac{x^2}{2} + \frac{x^3}{3}$

let $x = 0.01$

$$\frac{e^{0.01}}{1+0.01} \approx 1 + \frac{(0.01)^2}{2} + \frac{(0.01)^3}{3}$$

$$\frac{e^{0.01}}{1.01} \approx 1.0000496667$$

$\approx \mathbf{1.0000497}$ ***(to 7dp)***

VOICE OF THE PROFESSOR (VOP)

Key formulae:

Series expansion of e^x from MF7 formula booklet or using Maclaurin's theorem

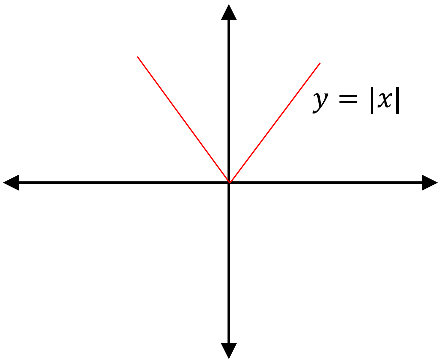
The universal binomial expansion

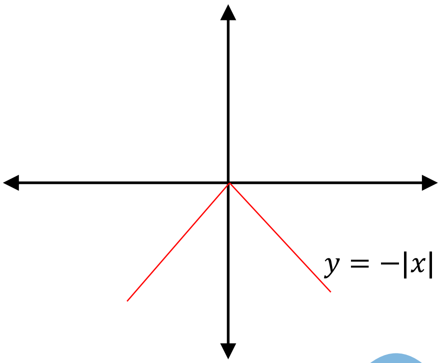
[Award 6 marks]

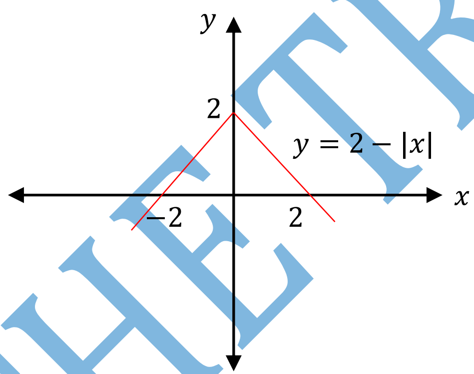
[Award 2 marks]

10. (a) $f(x) = 2 - |x|$ *and* $g(x) = \frac{1}{3}x + 1$

Stages in sketching the graph of $f(x)$







Stages in sketching the graph of $g(x)$
use a table of values

$$g(x) = \frac{1}{3}x + 1$$

x	0	-3
y	1	0

VOICE OF THE PROFESSOR (VOP)

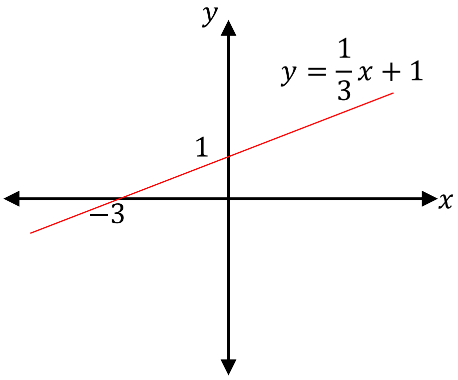
$y = |x|$ *is a reflection of the part of the graph of* $y = x$ *below the* $x-axis$*, mirror line* $y = 0$

$y = -|x|$: *reflection of the graph of* $y = |x|$*, mirror line* $y = 0$

$y = 2 - |x|$ *is a translation of* $y = -|x|$*, translation vector* $\binom{0}{2}$

OR

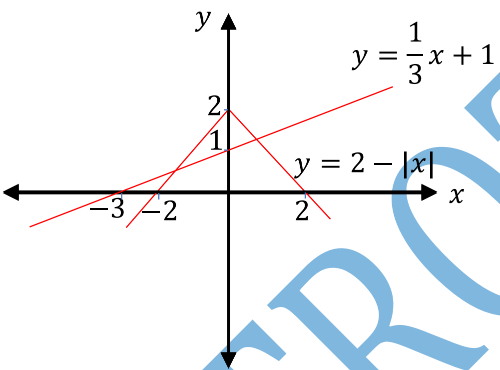
For the graph of $y = 2 - |x|$ *the first part is* $y = 2 - x$ *with* $-(ve)$ *grad and the other part is* $y = 2 + x$ *with* $+(ve)$ *grad*



VOICE OF THE PROFESSOR (VOP)

Could also use transformation of curves

$\therefore$ ***the required sketch is***



[Award 3 marks]

(b) $2 - |x| = \frac{1}{3}x + 1$

isolate the absolute value

$2 - 1 - \frac{1}{3}x = |x|$

$1 - \frac{1}{3}x = |x|$

square both sides

$\left(1 - \frac{1}{3}x\right)^2 = (x)^2$

bring all terms to one side with a bias towards differnce of 2 squares

$\left(1 - \frac{1}{3}x\right)^2 - (x)^2 = 0$

factorise using difference of 2 sqs

$\left(1 - \frac{1}{3}x + x\right)\left(1 - \frac{1}{3}x - x\right) = 0$

$\left(1 + \frac{2}{3}x\right)\left(1 - \frac{4}{3}x\right) = 0$

Accept ORA

$x = -\frac{3}{2}$ or $\frac{3}{4}$

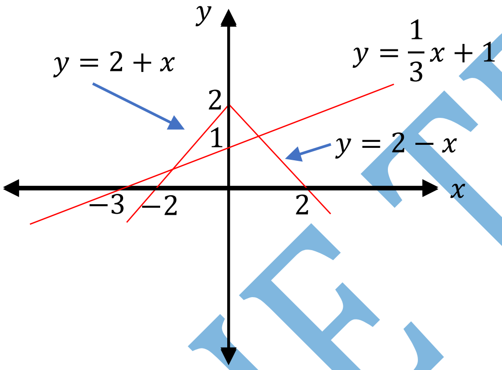
$y = 2 - \left|-\frac{3}{2}\right|$
$= 2 - \frac{3}{2}$
$= \frac{1}{2}$
OR
$y = 2 - \left|\frac{3}{4}\right|$
$y = 2 - \frac{3}{4}$
$y = \frac{5}{4}$

$\therefore$ coordinates of intersection are $\left(-\frac{3}{2}; \frac{1}{2}\right)$ and $\left(\frac{3}{4}; \frac{5}{4}\right)$

VOICE OF THE PROFESSOR (VOP)

[Award 5 marks]

Alternative solution: The graphical method



For the graph of $y = 2 - |x|$
$y = 2 - x$ (with $-(ve)$ grad and the other part is $y = 2 + x$ with $+(ve)$ grad

Finding the first coordinates

$2 + x = \frac{1}{3}x + 1$
$x - \frac{1}{3}x = 1 - 2$
$\frac{2}{3}x = -1$
$x = -\frac{3}{2}$

$\therefore y = 2 + \left(-\frac{3}{2}\right)$
$= 2 - \frac{3}{2}$
$= \frac{1}{2}$

VOICE OF THE PROFESSOR (VOP)

finding the second coordinates

$$2 - x = \frac{1}{3}x + 1$$
$$-x - \frac{1}{3}x = 1 - 2$$
$$-\frac{4}{3}x = -1$$
$$x = \frac{3}{4}$$

$$y = 2 - \frac{3}{4}$$
$$y = \frac{5}{4}$$

$\therefore$ ***coordinates of intersection are***

$$\left(-\frac{3}{2};\frac{1}{2}\right) \quad \textbf{and} \quad \left(\frac{3}{4};\frac{5}{4}\right)$$

[Award 5 marks]

11. (a) $\frac{2x}{(x+2)(x-2)(x-1)} = \frac{A}{x+2} + \frac{B}{x-2} + \frac{C}{x-1}$

Use the linear factor approach.

$$2x = A(x-2)(x-1) + B(x+2)(x-1) + C(x+2)(x-2)$$

$let\ x = 1$

$$2 = -3c$$
$$-\frac{2}{3} = C$$

Could also compare coefficients

$let\ x = 2$

$$4 = 4B$$
$$1 = B$$

$let\ x = -2$

$$-4 = 12A$$
$$-\frac{1}{3} = A$$

$$\frac{\mathbf{2x}}{\mathbf{(x+2)(x-2)(x-1)}} = \frac{\mathbf{-1}}{\mathbf{3(x+2)}} + \frac{\mathbf{1}}{\mathbf{x-2}} - \frac{\mathbf{2}}{\mathbf{3(x-1)}}$$

[Award 4 marks]

(b) $\frac{2x}{(x+2)(x-2)(x-1)} < 0$

Take the ranges which give negative result

critical values are

$x = -2, \quad x = 0, \quad x = 1, \quad x = 2$

Using table

	$x<-2$	$-2<x<0$	$0<x<1$	$1<x<2$	$x>2$
$2x$	$-$	$-$	$+$	$+$	$+$
$x+2$	$-$	$+$	$+$	$+$	$+$
$x-2$	$-$	$-$	$-$	$-$	$+$
$x-1$	$-$	$-$	$-$	$+$	$+$
$f(x)$	$+$	$-$	$+$	$-$	$+$

the solution set is

$$\mathbf{-2<x<0 \quad \cup \quad 1<x<2}$$

[Award 4 marks]

12. (a) $p=-4+3i \quad and \quad q=-1+\sqrt{3}i$

$$|p|=\sqrt{(-4)^2+3^2}$$
$$=\mathbf{5}$$
$$|q|=\sqrt{(-1)^2+\left(\sqrt{3}\right)^2}$$
$$=\mathbf{2}$$

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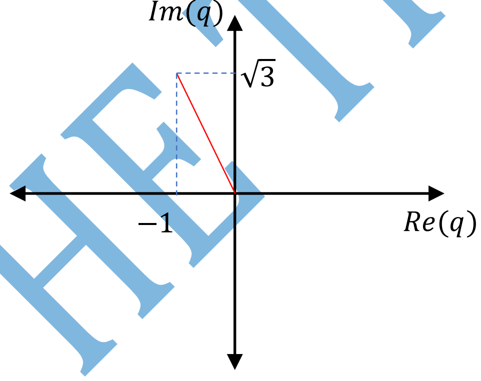
[Award 4 marks]

(b) (i) $\boldsymbol{Arg(q)=\theta=\tan^{-1}\left(\frac{\sqrt{3}}{1}\right)}$

$\boldsymbol{basic\ angle\ \alpha=\frac{\pi}{3}}$

sketch the Argand diagram

If $Z=a+ib$, $Arg(Z)=\tan^{-1}\left(\frac{b}{a}\right)$



Since our complex number q is the second quadrant ,
argument, $\theta=\pi-\alpha$

$$\therefore \theta=\pi-\frac{\pi}{3}$$
$$=\frac{\mathbf{2\pi}}{\mathbf{3}}$$

[Award 2 marks]

(b) (ii)

$$pq^2 = (-4+3i)\left(-1+\sqrt{3}i\right)^2$$
$$= (-4+3i)\left(1-2\sqrt{3}i+3i^2\right)$$
$$= (-4+3i)\left(1-2\sqrt{3}i-3\right)$$
$$= (-4+3i)\left(-2-2\sqrt{3}i\right)$$
$$= 8+8\sqrt{3}i-6i-6\sqrt{3}i^2$$
$$= 8+8\sqrt{3}i-6i+6\sqrt{3}$$
$$group$$
$$= 8+6\sqrt{3}+8\sqrt{3}i-6i$$
$$= \mathbf{\left(8+6\sqrt{3}\right)+\left(8\sqrt{3}-6\right)}\boldsymbol{i}$$

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$i^2 = -1$

[Award 3 marks]

(b) (iii)

$$\frac{-4+3i}{-1+\sqrt{3}i}\times\frac{-1-\sqrt{3}i}{-1-\sqrt{3}i}$$
$$= \frac{4+4\sqrt{3}i-3i-3\sqrt{3}i^2}{1+3i^2}$$
$$group$$
$$= \frac{4+3\sqrt{3}+4\sqrt{3}i-3i}{1+3}$$
$$= \frac{4+3\sqrt{3}+\left(4\sqrt{3}-3\right)i}{4}$$
$$break$$
$$= \mathbf{\frac{4+3\sqrt{3}}{4}+\frac{\left(4\sqrt{3}-3\right)\boldsymbol{i}}{4}}$$

$i^2 = -1$

[Award 3 marks]

13. $y = 10-\frac{5}{x}$.........(1)

$y+x = 6$..........(2)

(a) $substitute\ (1)\ \ in\ \ (2)$

$$10-\frac{5}{x}+x = 6$$
$$clear\ fraction$$
$$10x-5+x^2 = 6x$$
$$x^2+10x-6x-5 = 0$$
$$x^2+4x-5 = 0$$
$$x^2-x+5x-5 = 0$$
$$(x+5)(x-1) = 0$$
$$x = -5\ or\ x = 1$$

$$when\ x = -5$$
$$y-5 = 6$$
$$y = 11$$

$$when\ x = 1$$
$$y = 6-1$$
$$y = 5$$

$\therefore$ ***the coordinates of P & Q are*** $\mathbf{(-5;11)}$ ***and*** $\mathbf{(1;5)}$

[Award 5 marks]

(b) $Midpoint\ of\ PQ$

$let\ P = (-5; 11)\ \ and\ \ \ Q(1; 5)$

$midpoint\ of\ PQ = \left(\frac{-5+1}{2}; \frac{11+5}{2}\right)$

$\mathbf{(-2; 8)}$

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[2 marks]

(c) $we\ need\ gradient\ and\ a\ point$

$The\ point\ is\ (-2; 8)$

$finding\ gradient$

$$Grad\ of\ PQ = \frac{11-5}{-5-1}$$
$$= \frac{6}{-6}$$
$$= -1$$

$$Grad\ of\ perpendicular\ bisector = \frac{-1}{-1}$$
$$= 1$$

$now\ the\ equation\ of\ the\ line\ is$

$$y - y_1 = m(x - x_1)$$
$$y - 8 = 1(x - -2)$$
$$y - 8 = x + 2$$
$$\boldsymbol{y - x - 10 = 0}$$

[Award 3 marks]

14. (a) $y = x^2 + px + q \quad turning\ point\ (-1; -5)$

Key concept

$at\ turning\ point, \frac{dy}{dx} = 0$

$$\frac{dy}{dx} = 2x + p$$

$$at\ turning\ point, \frac{dy}{dx} = 0$$
$$\therefore 2x + p = 0$$

$$when\ x = -1$$
$$2(-1) + p = 0$$
$$p = 2$$

$substituting\ value\ of\ p, x\ and\ y\ in\ the\ original\ equation$

$$y = x^2 + px + q$$

$$-5 = (-1)^2 + 2(-1) + q$$

$$-5 = -1 + q$$

$$-4 = q$$

$$\boldsymbol{\therefore p = 2, \qquad q = -4}$$

[Award 5 marks]

(b) $The\ equation\ of\ the\ curve\ is\ y = x^2 + px + q$

$$y = x^2 + 2x - 4$$

$To\ find\ the\ equation\ of\ a\ tangent, we\ need\ gradient\ and\ a\ point$

$Finding\ the\ point$

$When\ the\ curve\ cuts\ the\ y-axis, x = 0$

$$y = 0^2 + 2(0) - 4$$
$$y = -4$$
$$the\ point\ is\ (0; -4)$$

$Finding\ gradient$

$$Grad = \frac{dy}{dx} = 2x + 2$$
$$at\ x = 0,$$
$$grad = 2$$

$now\ the\ equation\ of\ the\ tangent\ is$

$$y - y_1 = m(x - x_1)$$
$$y - -4 = 2(x - 0)$$
$$y + 4 = 2x$$
$$\mathbf{y - 2x + 4 = 0}$$

$To\ find\ the\ equation\ of\ a\ normal, we\ need\ gradient\ and\ a\ point$

$$The\ point\ is\ (0; 4)$$

$Finding\ gradient\ of\ normal$

$$grad = \frac{-1}{grad\ of\ tangent}$$
$$= \frac{-1}{2}$$

$now\ the\ equation\ of\ the\ normal\ is$

$$y - y_1 = m(x - x_1)$$
$$y - -4 = -\frac{1}{2}(x - 0)$$

$$2y + 8 = -x$$

$$\mathbf{2y + x + 8 = 0}$$

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Key concept

$grad\ of\ normal = \frac{-1}{grad\ of\ tangent}$

[Award 5 marks]

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15. (a) $Image = operator \times original$

$$\begin{pmatrix}7\\2\end{pmatrix} = \begin{pmatrix}1 & k\\0 & 1\end{pmatrix}\begin{pmatrix}1\\2\end{pmatrix}$$

$$\begin{pmatrix}7\\2\end{pmatrix} = \begin{pmatrix}1+k\\2\end{pmatrix}$$

$$\therefore 7 = 1 + k$$

$$7 - 1 = k$$

$$\mathbf{6 = k}$$

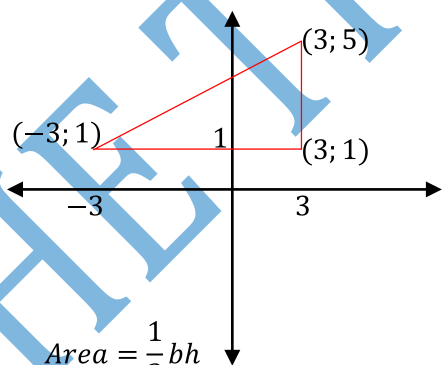
$$\therefore \textit{The shear factor is } 6$$

[Award 3 marks]

Points to note

- Shear is the only transformation with **one zero**.
- Shear **parallel to the x-axis means x-axis invariant or in the y-direction** henceon the operator, the x-direction **doesn't** have the shear factor.
- Hence the operator is $\begin{pmatrix}1 & k\\0 & 1\end{pmatrix}$
- However, do not confuse this with stretch. In stretch, parallel to the x-axis means y-axis invariant or in the x-direction

(b(i)) Method 1



$$Area = \frac{1}{2}bh$$

$$= \frac{1}{2}AB.BC$$

$$AB = \sqrt{(-3-3)^2 + (1-1)^2}$$

$$= 6$$

$$BC = \sqrt{(3-3)^2 + (5-1)^2}$$

$$= 4$$

$$\therefore Area = \frac{1}{2}bh$$

$$= \frac{1}{2} \times 6 \times 4$$

$$\mathbf{= 12\ units^2}$$

[Award 2 marks]

(b) (i)**Method 2: using vector cross product**

$we\ have\ the\ following\ position\ vectors$

$OA = \begin{pmatrix} -3 \\ 1 \end{pmatrix}$ $OB = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ $OC = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$

$Stage\ 1: Find\ the\ direction\ vectors$

$Nota\ Bene: Your\ direction\ vectors\ must\ start\ with\ the\ same\ letter$

$AB = OB - OA$

$= \begin{pmatrix} 3 \\ 1 \end{pmatrix} - \begin{pmatrix} -3 \\ 1 \end{pmatrix}$

$= \begin{pmatrix} 6 \\ 0 \end{pmatrix}$

$AC = OC - OA$

$= \begin{pmatrix} 3 \\ 5 \end{pmatrix} - \begin{pmatrix} -3 \\ 1 \end{pmatrix}$

$= \begin{pmatrix} 6 \\ 4 \end{pmatrix}$

$Stage\ 2: Use\ the\ formula\ Area\ of\ \Delta$

$= half\ \times the\ modulus\ of\ cross\ product$

$Area\ of\ \Delta ABC = \frac{1}{2}|AB \times AC|$

$= \frac{1}{2}| \begin{pmatrix} 6 \\ 0 \end{pmatrix} \times \begin{pmatrix} 6 \\ 4 \end{pmatrix}$

$= \frac{1}{2}(|6 \times 4 - (0 \times 6)|)$

$= \frac{1}{2} \times |24|$

$= \mathbf{12\ units^2}$

$=$

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[Award 2 marks]

(b)(i) **Method 3: Using vector cross product(The Direct Method)**

$we\ have\ the\ following\ position\ vectors$

$\boldsymbol{OA} = \begin{pmatrix} \mathbf{-3} \\ \mathbf{1} \end{pmatrix}$

$\boldsymbol{OB} = \begin{pmatrix} \mathbf{3} \\ \mathbf{1} \end{pmatrix}$ $\boldsymbol{OC} = \begin{pmatrix} \mathbf{3} \\ \mathbf{5} \end{pmatrix}$

$Stage\ 1: Inserting\ a\ factor\ of\ 1\ into\ each\ term\ we\ have$

$$\boldsymbol{The\ cross\ product} = \begin{vmatrix} \mathbf{1} & \mathbf{1} & \mathbf{1} \\ -3 & 3 & 3 \\ 1 & 1 & 5 \end{vmatrix}$$

$Stage\ 2: Use\ the\ formula\ Area\ of\ \Delta$

$= half\ \times the\ modulus\ of\ cross\ product$

$$Area\ of\ \Delta ABC = \frac{1}{2}\left\|\begin{vmatrix} \mathbf{1} & \mathbf{1} & \mathbf{1} \\ -3 & 3 & 3 \\ 1 & 1 & 5 \end{vmatrix}\right\|$$

Recall the 1st row of the sign change matrix

$+ \quad - \quad +$

$$= \frac{1}{2}[+(3\times 5 - 1\times 3) - (-3\times 5 - 1\times 3) + (-3\times 1 - 1\times 3)]$$

$$= \frac{1}{2}|(|12 + 18 - 6|)|$$

$$= \frac{1}{2}(|24|)$$

$$= \mathbf{12\ \ units^2}$$

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[Award 2 marks]

(b) (ii) $Image = operator \times original$

$$Image = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} -3 & 3 & 3 \\ 1 & 1 & 5 \end{pmatrix}$$

$$= \begin{pmatrix} (4\times -3 + 1\times 1) & (4\times 3 + 1\times 1) & (4\times 3 + 1\times 5) \\ (2\times -3 + 3\times 1) & (2\times 3 + 3\times 1) & (2\times 3 + 3\times 5) \end{pmatrix}$$

$$= \begin{pmatrix} -11 & 13 & 17 \\ -3 & 9 & 21 \end{pmatrix}$$

$\therefore$ ***the coordinates are***

$\boldsymbol{A,(-11;-3)}$

$\boldsymbol{B,(13;9)}$

$\boldsymbol{C,(17;21)}$

Image = operator × original

[Award 4 marks]

(b)(iii) *Use the formula area of image = determinant of operator × area of original*

$Area\ of\ \boldsymbol{A},B,\boldsymbol{C}, = \boldsymbol{detM} \times \boldsymbol{area\ of\ ABC}$

From item b(i), area of ABC = 12units²

Stage 1: Find the determinant

$$M = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$$

$$\det(M) = 4\times 3 - 2\times 1$$
$$= 10$$

Stage 2: Use the formula

$$Area\ of\ \boldsymbol{A},B,\boldsymbol{C}, = 10\times 12$$
$$= 120\ units^2$$

$Area\ of\ \boldsymbol{A},B,\boldsymbol{C}, = \boldsymbol{detM} \times \boldsymbol{area\ of\ ABC}$

[Award 2 marks]

16. (a) $\frac{dy}{dx} \propto 20 - y$

$$\frac{dy}{dx} = k(20 - y)$$

$when\ x = 0, \quad y = 0, \quad \frac{dy}{dx} = 1$

$$\therefore 1 = k(20 - 0)$$

$$\frac{1}{20} = k$$

now

$$\frac{dy}{dx} = \frac{1}{20}(20 - y)$$

$$\boldsymbol{\frac{dy}{dx} = 0.05(20 - y) \quad (Shown)}$$

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Substitute the given conditions to find k, the constant of variation.

[award 32 marks]

(b) $\frac{dy}{dx} = 0.05(20 - y)$

$$dy = 0.05(20 - y)dx$$

$$\frac{1}{20 - y}dy = 0.05dx$$

$$\int \frac{1}{20 - y}dy = \int 0.05dx$$

$$-\ln(20 - y) = 0.05x + c$$

$$\ln(20 - y) = -0.05x - c$$

$$e^{\ln(20-y)} = e^{-0.05x-c}$$

$$20 - y = e^{-0.05x} \cdot e^{-c}$$

$$let\ e^{c} = A$$

$$\therefore 20 - y = Ae^{-0.05x}$$

$$y = 20 - Ae^{-0.05x} \quad (the\ general\ solution)$$

$$when\ y = 0, \quad x = 0$$

$$0 = 20 - Ae^{-0.05(0)}$$

$$A = 20$$

$$\boldsymbol{\therefore y = 20 - 20e^{-0.05x}\ (the\ particular\ solution)}$$

Separate variables and integrate

[Award 6 marks]

(c) $y = 20 - 20e^{-0.05x}$

$$when\ x = 10$$

$$y = 20 - 20e^{-0.05(10)}$$

$$= 7.86938680$$

$$= 7.87$$

[Award 2 marks]

(d) $y = 20 - 20e^{-0.05(\infty)}$

$$y = 20 - 0$$

$$y = 20$$

As x becomes very large, y converges to 20

[Award 1 mark]

Dear Reader

CONSTRUCTIVE FEEDBACK ON THE LAYOUT AND ANY ERRORS AND OMISSIONS IS ALWAYS WELCOME

ENJOY YOUR STUDIES

BY T. HWATIRERA (THE TROTTER) 0774998145

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There is a strategy in everything we do.

We sink black! Siyaqeda umdlalo! Tinobatanidza masports!

MATHEW 19:26 But Jesus looked at them and said, "With men this is impossible, but with God all things are possible."